Taimour Mushtaq

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Career Overview

- Versatile AI/ML Engineer and Software Developer with 6+ months of combined internship experience spanning machine learning pipelines, deep learning, NLP, computer vision, and full-stack software development across Python, .NET C#, Java, and web technologies.
- Proven ability to deliver end-to-end solutions from data preprocessing and model training through evaluation, optimization, and Flask API deployment with a strong focus on clean code and reproducible workflows.
- Hands-on portfolio of 10+ projects demonstrating breadth across predictive modeling, healthcare AI, NLP classification, enterprise software systems, and database-driven applications.
- Effective collaborator in remote and hybrid team environments, with experience working on AI/ML tasks across multiple organizations simultaneously while managing independent project delivery.
- Seeking a paid AI/ML Internship or Junior Developer role where I can contribute immediately to real-world data-driven products while continuing to grow under senior mentorship.

Experience

Ezitech Institute *May 2025 - November 2025*

Machine Learning Intern | Rawalpindi, Pakistan

- Developed and evaluated supervised ML models (Logistic Regression, Random Forest, XGBoost, Gradient Boosting) using Scikit-learn on real-world structured datasets, applying cross-validation to ensure generalizability.
- Engineered comprehensive preprocessing pipelines including null-value imputation, outlier removal, Label & One-Hot Encoding, and Min-Max / Standard Scaler normalization, directly improving downstream model accuracy.
- Performed feature selection using correlation matrices, mutual information scores, and Recursive Feature Elimination (RFE), reducing feature space while retaining predictive signal.
- Tuned hyperparameters via GridSearchCV and RandomizedSearchCV, systematically optimizing key parameters (n_estimators, max_depth, C, alpha) to reduce overfitting and boost F1-scores on validation sets.
- Produced structured model evaluation reports using Accuracy, Precision, Recall, F1-score, MAE, and RMSE, presenting findings in Jupyter Notebooks with Matplotlib and Seaborn visualizations.

Developers Hub *Remote, Pakistan*

AI / ML Intern

- Designed and trained CNN-based image classification models in TensorFlow/Keras, applying data augmentation (rotation, flipping, zoom) and Dropout regularization to improve robustness on small datasets.
- Preprocessed image data with OpenCV resizing, grayscale conversion, histogram equalization, and noise filtering standardizing inputs for consistent model training across diverse image sources.
- Built NLP pipelines for text classification tasks: tokenization, stop-word removal, TF-IDF vectorization, and model training using Naive Bayes and Logistic Regression classifiers.
- Integrated trained ML/DL models into Flask REST APIs, exposing prediction endpoints consumed by a front-end interface, demonstrating practical AI application deployment skills.
- Maintained version-controlled project repositories on GitHub with structured README documentation, commit conventions, and modular code organization for team collaboration.

UMTI Tech *Remote*

AI / ML Intern | 1 Month

- Assisted in data collection, cleaning, and annotation pipelines for ML model training tasks in a fast-paced remote startup environment.

- Gained exposure to production-level ML workflow management, including experiment tracking and model versioning best practices.
- Collaborated asynchronously with senior engineers, adapting quickly to remote-first team communication and project handoff standards.

Archtech *Remote*

AI / ML Intern | 2 Months

- Contributed to AI/ML research and development tasks with a focus on model experimentation and evaluation for NLP and classification use cases.
- Implemented and tested multiple model architectures, documenting performance benchmarks and presenting findings to team leaders for decision-making.
- Participated in code reviews and sprint planning, gaining hands-on experience with agile workflows in a distributed development team.

Projects

Auto-Tagging System using LLM *Python | Transformers | HuggingFace | Flask*

- Built a zero-shot and few-shot tag generation system leveraging pre-trained Large Language Models (LLMs) via HuggingFace Transformers to automatically classify and tag unstructured text content without manually labeled training data.
- Designed a prompt engineering pipeline that structures input context and retrieves semantically relevant tags, enabling flexible deployment across different content domains with minimal configuration.
- Exposed the tagging engine as a Flask REST API endpoint, allowing seamless integration into content management workflows with JSON input/output handling.
- Evaluated output quality using precision/recall of generated tags against human-labeled ground truth, iterating on prompt structure to measurably improve tag relevance.

BERT News Article Classifier *Python | BERT | HuggingFace | PyTorch | Scikit-learn*

- Fine-tuned a pre-trained BERT (bert-base-uncased) model on a multi-class news classification dataset using HuggingFace Transformers and PyTorch, achieving strong accuracy across 4+ news categories.
- Implemented a full fine-tuning pipeline: tokenization with BertTokenizer, attention mask generation, DataLoader batching, AdamW optimizer with linear learning rate scheduling, and gradient clipping.
- Applied train/validation/test splits with stratification to ensure balanced class representation and avoid data leakage; monitored epoch-wise loss and accuracy curves to detect and address overfitting early.
- Benchmarked BERT against TF-IDF + Logistic Regression baseline, demonstrating a significant accuracy gain from transfer learning on limited labeled data.

Heart Disease Prediction System *Python | Scikit-learn | Pandas | Matplotlib | Flask*

- Developed a binary classification model to predict heart disease risk using the Cleveland Heart Disease dataset, comparing Logistic Regression, Random Forest, SVM, and KNN across 13 clinical features.
- Applied clinical feature engineering BMI-based risk bands, age grouping, and interaction terms between cholesterol and blood pressure improving model sensitivity (Recall) for the positive (at-risk) class.
- Prioritized Recall over Accuracy in model selection to minimize false negatives in a healthcare context where missing a high-risk patient has serious consequences.
- Deployed the best-performing model via a Flask web application with a clean HTML form interface, enabling real-time risk scoring from user-input clinical parameters.

Email Spam Detection - NLP Pipeline *Python | Scikit-learn | NLTK | TF-IDF*

- Engineered a complete NLP classification pipeline for binary email spam detection: text cleaning (HTML tag removal, punctuation stripping), tokenization, lemmatization, stop-word removal, and TF-IDF feature extraction.
- Trained and compared Multinomial Naive Bayes, Logistic Regression, and Linear SVM classifiers; achieved F1-score above 0.97 on the SpamAssassin dataset using the best-performing model.
- Applied class-weight balancing and threshold tuning to maximize Precision on spam predictions, minimizing false positives that would discard legitimate emails.

School Management System *C# | .NET | Windows Forms | SQL Server | ADO.NET*

- Designed and built a full-featured desktop School Management System covering student enrollment, attendance tracking, grade management, fee processing, and timetable scheduling for administrative staff.
- Architected a 3-tier application structure (Presentation / Business Logic / Data Access) using .NET Windows Forms and ADO.NET with parameterized SQL queries to prevent SQL injection vulnerabilities.
- Modeled a normalized relational database schema in SQL Server with 12+ tables (Students, Classes, Subjects, Grades, Fees, Teachers), enforcing referential integrity through foreign keys and constraints.
- Implemented role-based access control (Admin, Teacher, Staff) with login authentication and session management, ensuring data security and appropriate feature visibility per user role.

Customer Churn Prediction – End-to-End ML Pipeline *Python | Scikit-learn | XGBoost | Pandas*

- Built a production-style ML pipeline from raw CSV ingestion through feature preprocessing, SMOTE-based class balancing, model training, and evaluation, achieving high F1-score on the positive churn class.
- Compared Random Forest, XGBoost, and Gradient Boosting classifiers with cross-validated hyperparameter tuning, selecting the optimal model based on AUC-ROC and business-relevant Recall thresholds.
- Visualized feature importances, confusion matrices, and ROC-AUC curves to translate model outputs into actionable retention strategy insights.

Technical Skills

Programming Languages: Python, C#, C++, Java, SQL, HTML, CSS
Machine Learning: Supervised & Unsupervised Learning, Classification, Regression, Feature Engineering, Hyperparameter Tuning (GridSearchCV, RandomizedSearchCV), Cross-Validation, SMOTE
Deep Learning & NLP: CNNs, Neural Networks, BERT Fine-Tuning, TF-IDF, Tokenization, Transformers, HuggingFace, TensorFlow, Keras, PyTorch
Libraries: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, OpenCV, NLTK, Flask
Databases & Backend: SQL Server, ADO.NET, MySQL, .NET (Windows Forms), Flask REST APIs
Tools & Platforms: Git, GitHub, VS Code, Visual Studio Code, Jupyter Notebook, Google Collab, PyCharm, IntelliJ, Apache NetBeans
Soft Skills: Remote Collaboration, Technical Documentation, Problem-Solving, Fast Learning

Education

Federal Urdu University of Arts, Science & Technology *2024 - Expected 2028*
Bachelor of Science in Computer Science | Islamabad, Pakistan

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Object-Oriented Programming, Probability & Statistics, Discrete Mathematics, Artificial Intelligence

Certifications

AI / Machine Learning Course *2025 - 2026*
PNY Trainings, Rawalpindi | Machine Learning Fundamentals, Python for Data Science & AI, Developing AI Applications with Flask

Machine Learning & AI Specialization *2025*
Includes: Supervised & Unsupervised ML, Deep Learning Fundamentals, Model Deployment with Flask

Data Science Professional Courses *2025*
Data Analysis & Visualization with Python | Databases and SQL for Data Science